Embedded systems exp 29

```
#include <systemc.h>
SC_MODULE(priority_interrupt) {
 sc_event low_int, high_int;
 void low_ISR() {
 while(true) {
 wait(low_int);
 cout << "Low Priority Interrupt at "
 << sc_time_stamp() << endl;
 wait(3, SC_SEC);
 }
 }
 void high_ISR() {
 while(true) {
 wait(high_int);
 cout << "High Priority Interrupt at "
 << sc_time_stamp() << endl;
 wait(1, SC_SEC);
 }
 }
 void generator() {
 wait(1, SC_SEC);
 low_int.notify();
 wait(1, SC_SEC);
 high_int.notify();
 }
 SC_CTOR(priority_interrupt) {
 SC_THREAD(low_ISR);
 SC_THREAD(high_ISR);
```

```
 SC_THREAD(generator);
 }
};
int sc_main(int argc, char* argv[]) {
 priority_interrupt obj("Priority_INT");
 sc_start(10, SC_SEC);
 return 0;
}
```